

Agentic Bullwhip Experiment

Version 4 — Detailed Results Report

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1. Experiment Overview

This experiment investigates whether LLM-based agents operating a 3-tier supply chain (OEM → Ancillary → Component) amplify or dampen demand variability — the **bullwhip effect**. A 2×2 factorial design varies two independent factors: information treatment (blind vs. context) and model tier (lightweight: gpt-4.1-mini vs. reasoning: o1). Each of the four resulting configurations was run 5 times over 12 ordering periods, producing 20 complete experimental runs and 720 individual LLM calls.

Factor	Level A	Level B
Information treatment	Blind (no demand history)	Context (seasonal context + history)
Model tier	Lightweight (gpt-4.1-mini)	Reasoning (o1)

Supply chain structure: Three tiers operate in serial. Demand originates externally at the OEM from a real-world inspired synthetic CSV (Dec 2024 – Dec 2025, 13 periods). The OEM's order becomes the ancillary's demand; the ancillary's order becomes the component's demand.

Primary metric: OVAR (Order Variance Amplification Ratio) = $\text{Var}(\text{orders}) / \text{Var}(\text{demand received})$, computed per tier per run and averaged across the 5 runs per configuration. OVAR > 1 indicates bullwhip amplification; OVAR = 1 is perfect pass-through; OVAR < 1 indicates smoothing.

Control variables: Initial inventory 43,000 units at all tiers, 1-period lead time, order floor at 0 (no negative orders).

2. Data Quality & Run Integrity

All 20 runs completed without parse errors or undefined OVAR values. The dataset is clean.

Config	Runs	OVAR undefined	OEM CV%	Anc CV%	Comp CV%	Stability
blind lightweight	5/5	0	0.41	1.50	1.82	STABLE
blind reasoning	5/5	0	57.15	36.94	16.66	UNSTABLE
context lightweight	5/5	0	0.29	2.55	10.18	MOSTLY STABLE
context reasoning	5/5	0	22.86	32.76	25.10	UNSTABLE

CV > 10% flagged as high instability. o1 configurations exhibit wide run-to-run variance — conclusions drawn from their OVAR means must be treated as directionally informative, not statistically definitive, without larger sample sizes (n ≥ 20 recommended).

3. Primary Results — OVAR by Configuration

OVAR is the central experimental metric. The table below reports the mean ± standard deviation across 5 runs for each configuration-tier combination, along with the chain-average OVAR.

Config	Model	Treatm ent	OEM OVAR mean ± std	CV %	Ancillary OVAR mean ± std	CV %	Component OVAR mean ± std	CV %	Chain Avg
blind lightweight	gpt-4.1-mi ni	Blind	2.267 ± 0.009	0.4	2.938 ± 0.044	1.5	4.266 ± 0.078	1.8	3.157
blind reasoning	o1	Blind	4.200 ± 2.400	57.1	3.656 ± 1.350	36.9	3.649 ± 0.608	16.7	3.835

Config	Model	Treatm ent	OEM OVAR mean ± std	CV %	Ancillary OVAR mean ± std	CV %	Component OVAR mean ± std	CV %	Chain Avg
context lightweight	gpt-4.1-mi ni	Context	2.237 ± 0.006	0.3	3.138 ± 0.080	2.5	3.412 ± 0.347	10.2	2.929
context reasoning	o1	Context	6.349 ± 1.452	22.9	4.191 ± 1.373	32.8	2.698 ± 0.677	25.1	4.412

Green chain avg = best (lowest). Red = worst. CV% cells in red indicate high instability (>10%).

3.1 Chain-Average OVAR Ranking

Rank	Config	Chain Avg OVAR	Assessment
Best (lowest)	context lightweight	2.929	Context + lightweight is the best overall performer
2nd	blind lightweight	3.157	Blind + lightweight performs well
3rd	blind reasoning	3.835	Blind + reasoning is inconsistent but moderate
Worst (highest)	context reasoning	4.412	Context + reasoning is the worst — fully inverted cascade

4. Key Findings

4.1 All Configurations Confirm the Bullwhip Effect

Every OVAR value across all 4 configurations and 3 tiers is substantially above 1.0. LLM-based agents inherently amplify demand variability under the experimental conditions tested. The central question is not *whether* bullwhip occurs, but *how the cascade pattern differs across configurations*.

4.2 Blind Configs Reproduce the Textbook Monotone Cascade

Both blind configurations follow the classical pattern: OEM (~2.27) → Ancillary (~2.94–3.66) → Component (~3.65–4.27), with each upstream tier amplifying further. The lightweight and reasoning models produce similar magnitudes in the blind treatment, but the reasoning model shows much higher run-to-run instability (OEM CV 57.15% vs 0.41%). Model power alone does not reduce bullwhip without additional context.

4.3 Context + Lightweight Partially Dampens the Cascade

Context_lightweight achieves the best chain-average OVAR (2.929) and best pattern detection (elevation score 0.833). Downstream tiers benefit most: component OVAR drops from 4.266 to 3.412 (–20%) vs blind_lightweight, while OEM remains virtually unchanged (2.267 → 2.237). The improvement is consistent across runs (all CVs ≤ 10.18%). At \$0.0068/run, this is the most cost-effective configuration.

4.4 Context + Reasoning Creates an Anomalous Inverted Cascade

The most striking finding of this experiment. Context_reasoning produces a **fully inverted cascade**: OEM OVAR (6.349) > Ancillary (4.191) > Component (2.698). The top of the chain is the worst amplifier; the bottom is the best. This is the opposite of all other configurations and of classical bullwhip theory.

Config	OEM OVAR	Ancillary OVAR	Component OVAR	Cascade Direction
blind_lightweight	2.267	2.938	4.266	Normal ↑ upstream
context_lightweight	2.237	3.138	3.412	Normal ↑ upstream
blind_reasoning	4.200	3.656	3.649	Flat/mild reversal
context_reasoning	6.349	4.191	2.698	FULLY INVERTED ↓ upstream

Mechanistic explanation: o1's extended chain-of-thought reasoning processes the seasonal context prompt differently depending on tier position. At OEM (which sees raw external demand + seasonal annotations), o1 constructs elaborate anticipatory orders before expected festival peaks — *variance injection*. At Component (which receives already-distorted upstream orders), o1 with context can recognise that large orders are festival-driven spikes and moderates its response — *spike-recognition dampening*. gpt-4.1-mini does not exhibit this reversal because it applies context more uniformly and with less elaboration.

5. Hypothesis Verdicts

Hypothesis	Prediction	Result	Verdict
H1	Context OVAR < Blind OVAR at all three tiers	Context raises OEM OVAR (+1.059 avg across models); only improves component	REJECTED

Hyp oth esis	Prediction	Result	Verdict
H2	Blind-reasoning ≈ blind-lightweight (model doesn't matter)	Blind-reasoning OEM OVAR 4.200 vs 2.267 for blind-lightweight — delta 1.933, CV 57%	REJECTED
H3	Context-reasoning achieves the lowest chain OVAR	Context-reasoning achieves the HIGHEST chain OVAR (4.412) — worst in experiment	REJECTED
H4	Context agents score higher on seasonal pattern detection	True for context-lightweight (0.833 vs 0.700 elevation score); reversed for context-reasoning (0.600)	PARTIALLY SUPPORTED

All three primary hypotheses are rejected. The data do not support the intuition that more information (context) and more reasoning capability (o1) jointly improve supply chain ordering behaviour as measured by OVAR.

6. Main Effects & Context × Model Interaction

6.1 Context Main Effect (averaged across both models)

Tier	Blind mean OVAR	Context mean OVAR	Delta	Direction
OEM	3.233	4.293	+1.059	Context WORSE
Ancillary	3.297	3.664	+0.367	Context WORSE
Component	3.958	3.055	−0.903	Context BETTER

6.2 Model Main Effect (averaged across both treatments)

Tier	Lightweight mean OVAR	Reasoning mean OVAR	Delta	Direction
OEM	2.252	5.274	+3.022	Reasoning WORSE
Ancillary	3.038	3.923	+0.885	Reasoning WORSE
Component	3.839	3.174	−0.665	Reasoning BETTER

6.3 Interaction: Context × Model at OEM Tier

The most important structural finding is a significant interaction between context and model at the OEM tier. The context effect *reverses sign* depending on the model:

Tier	Context effect on gpt-4.1-mini	Context effect on o1	Interaction
OEM	−0.030 (marginal improvement)	+2.149 (substantial worsening)	STRONG REVERSAL
Ancillary	+0.199 (slight worsening)	+0.535 (worsening)	Consistent
Component	−0.855 (improvement)	−0.952 (improvement)	Consistent

At the component tier, context improves OVAR for both models by similar magnitudes. At the OEM tier, context is effectively neutral for gpt-4.1-mini and severely damaging for o1. **Design recommendation: context provision and model selection cannot be treated as independent choices.**

7. Stability Analysis (Coefficient of Variation)

Run-to-run stability is critical for operational deployability. A high CV means the same configuration can produce qualitatively different outcomes in successive months — which is unacceptable for inventory planning.

Config	OEM CV%	Ancillary CV%	Component CV%	Max CV%	Assessment
blind lightweight	0.4	1.5	1.8	1.8	STABLE
blind reasoning	57.1	36.9	16.7	57.1	UNSTABLE
context lightweight	0.3	2.5	10.2	10.2	MOSTLY STABLE
context reasoning	22.9	32.8	25.1	32.8	UNSTABLE

Lightweight model runs are essentially deterministic (all CVs $\leq 10.18\%$). o1 runs are highly stochastic: blind_reasoning OEM CV of 57.15% means the 95% CI for the true mean spans approximately [2.0, 6.4] —

overlapping substantially with context_reasoning OEM (6.349). With n=5 runs, o1 conclusions are directionally informative only.

8. Secondary Metrics

8.1 Stockouts

Stockout counts (periods where backlog > 0) decrease in context_reasoning, but this is a consequence of massive over-ordering rather than improved planning efficiency.

Config	OEM Stockouts	Ancillary Stockouts	Component Stockouts
blind lightweight	8.0	6.0	7.4
blind reasoning	7.4	5.8	6.8
context lightweight	8.0	6.0	5.6
context reasoning	5.4	2.8	4.6

8.2 Excess Inventory (end-of-simulation, mean across runs)

Config	OEM Excess	Ancillary Excess	Component Excess	Chain Total
blind lightweight	0	50,706	58,654	109,360
blind reasoning	4,659	103,229	222,761	330,649
context lightweight	0	70,748	80,498	151,246
context reasoning	14,527	268,901	371,300	654,728

Context_reasoning accumulates 654,728 units of chain-wide excess inventory — approximately 6× blind_lightweight. This is the physical correlate of high OEM/ancillary OVAR: large anticipatory orders fill pipelines even when demand does not materialise as expected.

8.3 Total Units Ordered (12 periods, mean across 5 runs)

Config	OEM Total	Ancillary Total	Component Total	Chain Total
blind lightweight	570,567	578,273	583,860	1,732,700
blind reasoning	570,627	588,930	624,777	1,784,334
context lightweight	570,639	598,387	626,744	1,795,770
context reasoning	594,470	651,563	698,960	1,944,993

Context_reasoning orders 33% more component units than blind_lightweight (698,960 vs 583,860). A modest 4.2% over-order at OEM cascades into 80,996 additional ancillary units and 115,100 additional component units — the bullwhip cascade made physically concrete.

8.4 Peak Overshoot (max order / max demand)

Config	OEM Peak Overshoot	Ancillary Peak Overshoot	Component Peak Overshoot
blind lightweight	1.2239	1.4022	1.5393
blind reasoning	1.2704	1.5748	1.6067
context lightweight	1.2208	1.4019	1.5399

Config	OEM Peak Overshoot	Ancillary Peak Overshoot	Component Peak Overshoot
context reasoning	1.3669	1.6027	1.4722

9. Pattern Detection

Pattern score v2 = mean(keyword_score, elevation_score). Event periods are 3, 10, 11, 12. Keyword score measures seasonal vocabulary use; elevation score measures whether orders at event periods are ≥ 110% of the non-event baseline.

Config	Pattern Score mean ± std	CV%	Keyword Score mean ± std	Elevation Score mean ± std	Elev. CV%
blind lightweight	0.350 ± 0.020	5.8	0.000 ± 0.000	0.700 ± 0.041	5.8
blind reasoning	0.369 ± 0.033	8.9	0.037 ± 0.031	0.700 ± 0.041	5.8
context lightweight	0.417 ± 0.037	9.0	0.000 ± 0.000	0.833 ± 0.075	9.0
context reasoning	0.312 ± 0.095	30.3	0.025 ± 0.031	0.600 ± 0.200	33.3

Keyword score is zero for both lightweight configurations. gpt-4.1-mini agents never verbalised seasonal reasoning (no dasara, diwali, festive, budget, monsoon, fy-end keywords appeared). Both o1 configurations show occasional keyword use (blind: 0.038, context: 0.025), but inconsistently. Critically, gpt-4.1-mini can incorporate context into ordering decisions (shown by improved elevation score with context) without verbalising the seasonal reasoning — the behaviour changes but the language does not.

Context_lightweight achieves the best elevation score (0.833) — the only configuration to consistently elevate orders at event periods. Context_reasoning achieves the worst elevation score (0.600) with CV 33.33%, suggesting o1 sometimes over-rationalises and concludes that elevation is not warranted even when context implies it.

10. Cost Analysis

Cost is a tracking variable, not an experimental factor. It is recorded to characterise the resource footprint of each configuration. All cost figures are in USD.

Config	Model	Cost / run (USD)	Total 5 runs (USD)	Relative cost
blind lightweight	gpt-4.1-mini	\$0.0046	\$0.02	1×
blind reasoning	o1	\$1.6841	\$8.42	366×
context lightweight	gpt-4.1-mini	\$0.0068	\$0.03	1×
context reasoning	o1	\$3.5268	\$17.63	767×

Grand total experiment cost: \$26.11

o1 in the blind condition costs 366× more per run than gpt-4.1-mini with no measurable OVAR improvement (blind_reasoning chain avg 3.835 vs blind_lightweight 3.157). Context_reasoning is 767× more expensive than blind_lightweight and produces the worst chain-level OVAR. There is no dimension on which context_reasoning dominates blind_lightweight.

10.1 Cost-Performance Head-to-Head: context_reasoning vs blind_lightweight

Metric	blind_lightweight	context_reasoning	Winner
Chain avg OVAR	3.157	4.412	blind_lightweight
OEM OVAR	2.267	6.349	blind_lightweight
Component OVAR	4.266	2.698	context_reasoning
Chain excess inventory	109,360	654,728	blind_lightweight
Max CV%	1.82%	32.76%	blind_lightweight
Pattern score	0.350	0.313	blind_lightweight
Elevation score	0.700	0.600	blind_lightweight
Cost per run	\$0.0046	\$3.527	blind_lightweight

11. Recommendations

R1: Do not deploy o1 without hard order-change constraints

Blind_reasoning OEM CV of 57.15% and context_reasoning OEM CV of 22.86% indicate that o1 produces qualitatively different ordering strategies across identical runs. Deploying o1 in a live supply chain without clamping the maximum period-over-period order change (e.g., $\pm 20\%$ of rolling average) is operationally risky. The next experiment version should introduce order-smoothing as a third factor and test whether it restores o1 reliability without eliminating its component-tier benefit.

R2: Limit context provision to the component tier only

The context effect is beneficial and consistent at the component tier for both models ($\Delta \approx -0.90$ OVAR). At OEM, context is neutral for gpt-4.1-mini and severely damaging for o1. A targeted architecture where only the component agent receives seasonal context — while OEM and ancillary remain blind — may capture the downstream benefit without triggering upstream variance injection. This asymmetric information architecture should be tested as a new treatment.

R3: Increase n_runs to ≥ 20 for o1 configurations

With $n=5$ and CVs above 30%, the current experiment cannot determine whether blind_reasoning or context_reasoning OEM OVAR distributions meaningfully differ. Increasing to $n=20$ reduces the standard error of the mean by a factor of 2 and enables overlap quantification. Additionally, report the 10th, 50th, and 90th percentile OVAR per configuration to communicate operational risk more clearly than $\text{mean} \pm \text{std}$ alone.

R4: Fix gpt-4.1-mini JSON parse errors

gpt-4.1-mini occasionally formats numbers with comma separators (e.g., 27,959 instead of 27959), causing JSON parse failures that default order_quantity to 0. This suppresses some orders and slightly deflates OVAR for lightweight configs. Using JSON mode or a more explicit number-format instruction in the prompt should eliminate this artefact.

R5: Investigate context_reasoning's anti-pattern behaviour

Context_reasoning achieves the worst elevation score (0.600) despite having access to seasonal context. Examining raw o1 reasoning chains at event periods would reveal whether the model explicitly rejects seasonal anticipation or simply ignores the context. This is important for understanding whether an anti-overreaction instruction can constrain o1 at OEM without removing its component-tier benefit.

12. Limitations

Limitation	Detail
Small sample size for o1 (n=5)	CVs of 22–57% with $n=5$ produce wide confidence intervals. o1 conclusions are directionally informative but not statistically conclusive without $n \geq 20$ runs.
Single-period lead time	A lead time of 1 period is shorter than most real supply chains. Longer lead times typically amplify bullwhip effects and may change the relative ordering of configurations.
Homogeneous initial inventory	All tiers start at 43,000 units. Real supply chains have differentiated starting positions that may advantage or disadvantage certain agent strategies.
Pattern score as a proxy	Elevation score measures order magnitude at event periods, not whether the agent correctly attributes elevation to seasonal context. An agent that over-orders randomly at all periods would score identically to one that correctly identifies festival peaks.
No order smoothing constraints	The experiment applies only an order floor of 0. Real procurement systems impose rate-of-change constraints that would mechanically limit OVAR. Results represent unconstrained agent behaviour.

13. Summary Verdict

Verdict	Configuration	Rationale
Best bullwhip reduction	context_lightweight	Lowest chain avg OVAR (2.929), best elevation score (0.833), consistent across runs (max CV 10%), \$0.0068/run
Most dangerous combination	context_reasoning	Fully inverted cascade, OEM OVAR 6.349, 6× excess inventory, highest instability (CV 32.76%), \$3.527/run
Most consistent behaviour	blind_lightweight	All CVs ≤ 1.82%, predictable across runs, dominates context_reasoning on all metrics, \$0.0046/run
Most cost-efficient reasoning	blind_reasoning	o1 in blind mode costs 366× more than blind_lightweight with no statistically reliable OVAR improvement

The experiment's central conclusion is that **more capability does not monotonically improve outcomes**. The combination of richer context and stronger reasoning (context_reasoning) produces the worst chain-level result. The simpler, cheaper configuration (context_lightweight) produces the best. This is not a failure of reasoning models per se, but a demonstration that LLM agents with strong reasoning capabilities can construct elaborate, internally consistent justifications for over-ordering — and that unconstrained reasoning in information-rich environments amplifies rather than dampens supply chain instability.

Appendix — Raw Aggregated Metrics

Full per-tier means for all secondary metrics across all four configurations.

Config	Tier	OVAR mean	OVAR std	CV%	Stockouts	Excess Inv.	Total Ordered	Peak OShoot
blind lightweight	OEM	2.267	0.009	0.4	8.0	0	570,567	1.2239
	Ancillary	2.938	0.044	1.5	6.0	50,706	578,273	1.4022
	Component	4.266	0.078	1.8	7.4	58,654	583,860	1.5393
blind reasoning	OEM	4.200	2.400	57.1	7.4	4,659	570,627	1.2704
	Ancillary	3.656	1.350	36.9	5.8	103,229	588,930	1.5748
	Component	3.649	0.608	16.7	6.8	222,761	624,777	1.6067
context lightweight	OEM	2.237	0.006	0.3	8.0	0	570,639	1.2208
	Ancillary	3.138	0.080	2.5	6.0	70,748	598,387	1.4019
	Component	3.412	0.347	10.2	5.6	80,498	626,744	1.5399
context reasoning	OEM	6.349	1.452	22.9	5.4	14,527	594,470	1.3669
	Ancillary	4.191	1.373	32.8	2.8	268,901	651,563	1.6027
	Component	2.698	0.677	25.1	4.6	371,300	698,960	1.4722

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